



Muhammad Hanzla

DATA SCIENTIST · ML ENGINEER

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CORE SKILLS

Data Science

Python · Feature Engineering
Model Deployment

Machine Learning

Machine Learning · Deep Learning

Data Analysis

SQL & EDA · Data Visualisation
Tableau · Power BI

TOOLS & LIBRARIES

Pandas · NumPy · Scikit-learn
TensorFlow · PyTorch
XGBoost · SHAP · Plotly
Seaborn · Flask · FastAPI
Docker · Tableau · Power BI
HuggingFace · NLTK
Git/GitHub · Jupyter · MySQL
AWS/GCP

SOFT SKILLS

Problem Solving · Critical Thinking
Time Management · Adaptability
Attention to Detail · Self-Learning
Communication
Team Collaboration

LANGUAGES

Urdu (Native) · English (Fluent)

Muhammad Hanzla

DATA SCIENTIST

ML ENGINEER

DATA ANALYST

BS SOFTWARE ENGINEERING

PROFESSIONAL SUMMARY

Results-driven Data Scientist, ML Engineer and BS Software Engineering student (CGPA 3.83/4.0) with end-to-end expertise in building ML pipelines — from raw data wrangling to production deployment. Adept at translating complex datasets into actionable business insights through rigorous data storytelling and predictive modeling. Active Kaggle contributor with 6 published projects spanning classification, computer vision, NLP and time-series forecasting.

EDUCATION

BS Software Engineering

2024 — Present

NATIONAL UNIVERSITY OF MODERN LANGUAGES ISLAMABAD (FSD CAMPUS) — FAISALABAD, PAKISTAN

Specialisation: Data Science, Machine Learning & AI · Coursework: Algorithms, Statistics, Database Systems, AI, OOP, Programming

CGPA: 3.83 / 4.0

Intermediate (FSc Pre-Medical)

2021 — 2023

PUNJAB GROUP OF COLLEGES, SARGODHA, PAKISTAN

Matriculation (Science)

2019 — 2021

HIRA MODEL HIGH SCHOOL, SARGODHA, PAKISTAN

FEATURED PROJECTS

Scene Classifier — ResNet-50

Accuracy: 96% · Top-2: 99%

Transfer learning on 25K images, Grad-CAM + Docker + FastAPI

ResNet-50 · Grad-CAM · Docker

Sentiment Analysis — NLP

F1: 0.91 · Accuracy: 92%

TF-IDF → BiLSTM → DistilBERT fine-tuning on 50K IMDB reviews

BERT · BiLSTM · FastAPI

Fraud Detection System

AUC: 0.992 · AP: 0.88

284K transactions, 0.17% fraud · XGBoost + SHAP explainability

XGBoost · SHAP · IsoForest

Customer Churn Prediction

AUC: 0.97 · Accuracy: 94%

EDA → SMOTE → GBM → Flask REST API for real-time prediction

Python · XGBoost · SMOTE
Flask

House Price Prediction

RMSLE: 0.11 · R²: 0.89

40+ features, Ridge/Lasso/GBM stacking ensemble on Ames dataset

Pandas · Ridge/Lasso
Stacking

COVID-19 EDA Dashboard

50+ Visualizations · 90-Day Forecast

Global trends, wave analysis, 3 Plotly dashboards + Prophet forecast

Plotly · Prophet · Seaborn

AREAS OF EXPERTISE

- ML Pipelines:** End-to-end from EDA to deployment
- NLP:** BERT, TF-IDF, sentiment analysis, text classification
- Anomaly Detection:** Isolation Forest, SMOTE, imbalanced data
- Visualisation:** Plotly, Matplotlib, Seaborn
- Data Analysis:** EDA, statistical modeling, insights extraction
- Deep Learning:** CNN, LSTM, Transformers, Transfer Learning
- Data Wrangling:** Pandas, SQL, feature engineering, ETL
- Deployment:** Flask, FastAPI, Docker, REST APIs, Cloud
- Explainability:** SHAP, Grad-CAM, model interpretability
- Business Intelligence:** Tableau, Power BI, interactive dashboards

RESEARCH INTERESTS

- Explainable AI (XAI) — Interpretable machine learning models for transparent, trustworthy decision-making in real-world applications
- Computer Vision — Deep learning for object detection and segmentation
- Natural Language Processing — Large Language Models, transformer architectures & applied NLP for low-resource languages

CERTIFICATIONS & ACHIEVEMENTS



Kaggle Notebook Expert

Kaggle · 15+ published notebooks · Active data science community contributor



Data Science & Machine Learning Specialisation

Codanics [2025] · Comprehensive ML/DL curriculum with hands-on projects



Data Science Portfolio — 6 End-to-End Projects

muhammadhanzlaaltaf.github.io/portfolio · github.com/muhammadhanzlaaltaf